Block practice · More reproduction and development

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Describe labour and the advantages of breastfeeding

Describe fertilisation and early embryo development, including twins

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Practice workbook · Describe labour and the advantages of breastfeeding

Uterine contractions, oxytocin during birth and benefits of breast milk.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-DESCRIBE-BIRTH-AND-BREASTFEEDING

1. Question 1 · part b.i · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part b.i only.

1 (a) The words in the box give the names of some of the parts of the male and female reproductive systems.

epididymis	ovary	oviduct	seminal vesicle
testes	uterus	urethra	vagina
vulva			

Complete the table by giving the name of the part that matches the description.

(7)

Description	Name of part
duct that carries urine and sperm	
produces sperm	
where fertilised ovum is implanted	
produces the fluid part of semen	
where sperm is deposited during intercourse	
where fertilisation happens	
releases ova	

(b) Oxytocin and LH (luteinising hormone) are two hormones found in females.

(i) Describe the functions of oxytocin.

(3)

(ii) Describe the functions of LH.

(2)

(Total for Question 1 = 12 marks)

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1. Question 1 · part b.i · 3 marks · Working and answer

Working and answer

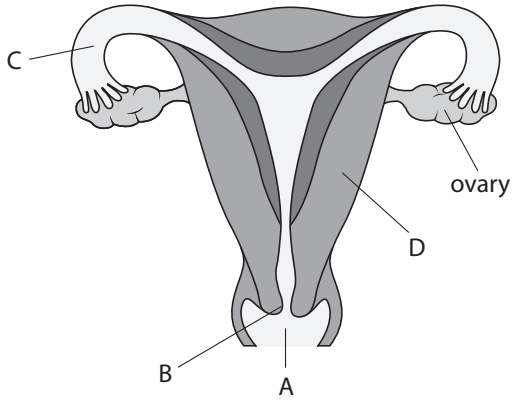
OLevels.com SCHOOL

2. Question 2 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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2. Question 2 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

OLevels.com SCHOOL

(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

.....

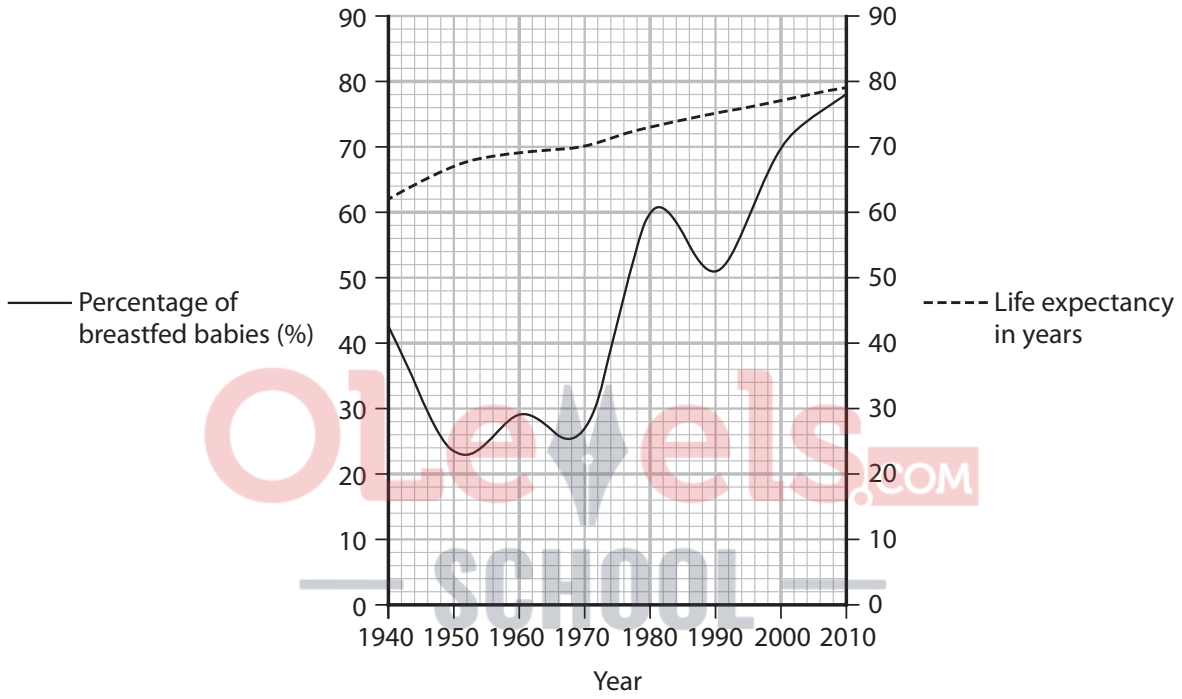
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2. Question 2 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

(d) The graph shows the percentage of babies that are breastfed and the life expectancy of breastfed babies from 1940 to 2010 in the UK.



(i) Compare the percentage of babies that were breastfed in 1940 with the percentage of babies that were breastfed in 2010.

(1)

(ii) The graph suggests that breastfeeding has no effect on life expectancy.

Describe how the graph shows this.

(2)

(Total for Question 2 = 10 marks)

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2. Question 2 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part g · 1 mark

June 2023 Paper 2R

Further practice · Attempt part g only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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3. Question 6 · part g · 1 mark

June 2023 Paper 2R

Further practice · Attempt part g only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

OLevels.com SCHOOL

(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

3. Question 6 · part g · 1 mark

June 2023 Paper 2R

Further practice · Attempt part g only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

(2)

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

(2)

OLevels.com SCHOOL

(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

(1)

(g) Which change occurs during the birth process? (line 20)

(1)

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

(Total for Question 6 = 13 marks)

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3. Question 6 · part g · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part d · 2 marks

June 2024 Paper 1R

Further practice · Attempt part d only.

6 (a) The table gives the names and functions of hormones in the menstrual cycle.

Complete the table by placing one tick (✓) in each row of the table to show the main function of each hormone in the menstrual cycle.

(3)

Hormone	Function of hormone			
	causes ovulation	builds up uterus lining after menstruation	maintains uterus lining during pregnancy	stimulates a follicle to mature
progesterone				
FSH				
oestrogen				

(b) The chance of pregnancy can be decreased by using contraception.

State **one** method of barrier contraception.

(1)

(c) In-vitro fertilisation (IVF) can be used to increase the chances of pregnancy.

Describe the process of IVF.

(4)

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4. Question 6 · part d · 2 marks

June 2024 Paper 1R

Further practice · Attempt part d only.

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(d) Many mothers decide to breastfeed their new-born baby.

Describe the advantages of breastfeeding.

(2)

(Total for Question 6 = 10 marks)

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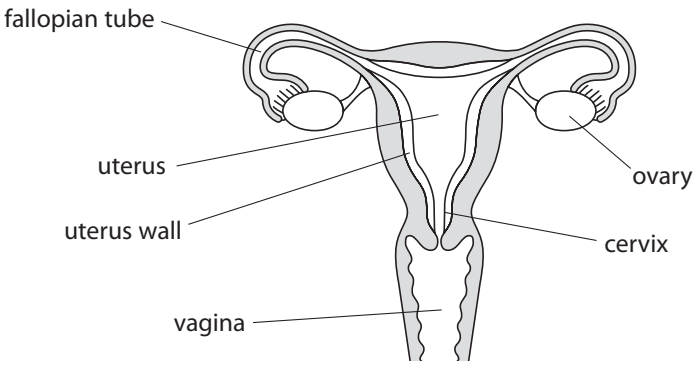
Working and answer

5. Question 2 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			
ovary			
vagina			

(ii) Describe the action of the muscles in the uterus wall during labour.

(2)

.....

.....

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P 7 4 4 9 5 A 0 6 2 8

5. Question 2 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

(1)

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

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5. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Describe fertilisation and early embryo development, including twins

From egg release through gamete fusion to mitotic division of the zygote, including how identical twins arise.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SEQUENCE-FERTILISATION-AND-EARLY-EMBRYO

1. Question 6 · part b · 4 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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1. Question 6 · part b · 4 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

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(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

1. Question 6 · part b · 4 marks · Working and answer

Working and answer

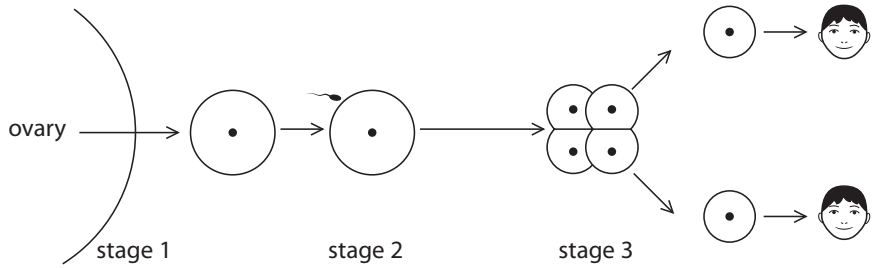
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2. Question 5 · part a.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.i only.

5 (a) The diagram shows how genetically identical twins are produced.



(i) Name the process that occurs at stage 1.

(1)

(ii) Name the process that occurs at stage 2.

(1)

(iii) Use information from the diagram to explain how genetically identical twins are formed.

(2)

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2. Question 5 · part a.i · 1 mark · Working and answer

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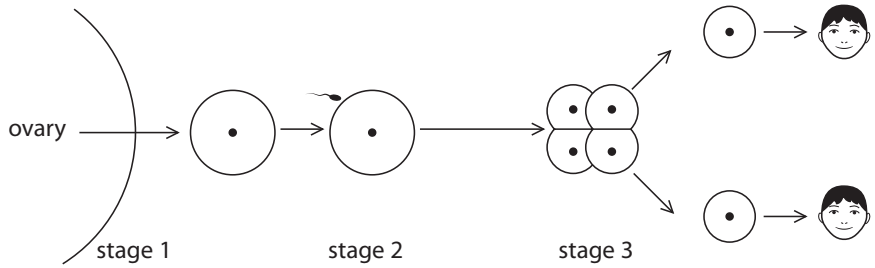
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3. Question 5 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

5 (a) The diagram shows how genetically identical twins are produced.



(i) Name the process that occurs at stage 1.

(1)

(ii) Name the process that occurs at stage 2.

(1)

(iii) Use information from the diagram to explain how genetically identical twins are formed.

(2)

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3. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

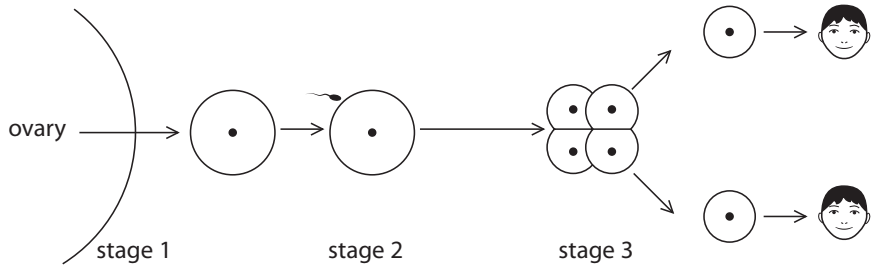
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4. Question 5 · part a.iii · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

5 (a) The diagram shows how genetically identical twins are produced.



(i) Name the process that occurs at stage 1.

(1)

(ii) Name the process that occurs at stage 2.

(1)

(iii) Use information from the diagram to explain how genetically identical twins are formed.

(2)

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4. Question 5 · part a.iii · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

(b) (i) In-vitro fertilisation (IVF) increases the chance of producing twins.

Which hormones are given to a female at the start of IVF treatment?

(1)

- ☒ **A** FSH and LH
- ☒ **B** LH and oestrogen
- ☒ **C** oestrogen and progesterone
- ☒ **D** progesterone and FSH

(ii) Describe the procedures in IVF after hormone treatment.

(4)

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(Total for Question 5 = 9 marks)

4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

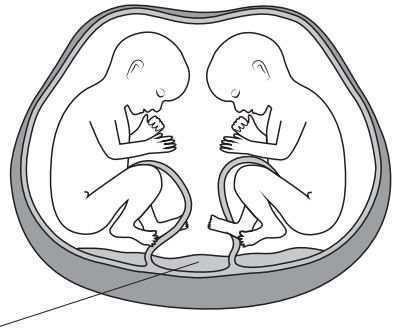
Further practice · Attempt part b.ii only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

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(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.



(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

(1)

percentile = ...

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5. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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OLevels.com SCHOOL

6. Question 6 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

OLevels.com SCHOOL

(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

6. Question 6 · part a · 2 marks · Working and answer

Working and answer

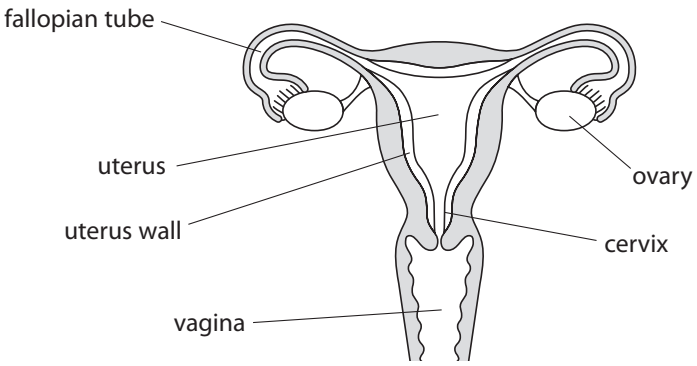
OLevels.com SCHOOL

7. Question 2 · part a.iii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			
ovary			
vagina			

(ii) Describe the action of the muscles in the uterus wall during labour.

(2)

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7. Question 2 · part a.iii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

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(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

(1)

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

OLevels.com SCHOOL

7. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL